

RM-Class-15-16



- **Class-15: Analysis of Variance Techniques**
- **Class-16: Non-parametric Tests**

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Class-15: Analysis of Variance Techniques

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Learning Objective Class 15

By the end of the class, a student should be able to answer the following Questions

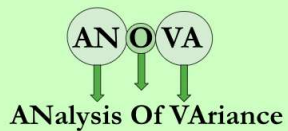
- What is ANOVA?
- State the procedure and example to illustrate the completely randomized design in a one-way ANOVA?
- State the procedure and example to illustrate the completely randomized design in a two-way ANOVA?
- What is a factorial design?
- Define Latin square design.

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What is ANOVA?

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ANOVA: Definition and Main Types



ANOVA (ANalysis Of VAriance) is a statistical test used to check if the mean values of three or more groups are significantly different from each other.

There are two types of ANOVA:

- **One-Way ANOVA:** Tests the **effect of one categorical independent variable** on a **single dependent variable**. For example, testing plant growth using three different fertilizers.
- **Two-Way ANOVA:** Tests the **effect of two independent categorical variables** on a **single dependent variable, plus how they interact**. For example, testing blood pressure changes based on drug type and gender.

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State the procedure and example to illustrate the completely randomized design in a one-way ANOVA?

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Key Concepts and When to Use One-Way ANOVA

Some of the Key Concepts of One-Way ANOVA are:

- **Factor:** the one thing we are changing or comparing (e.g., teaching method).
- **Levels:** the different groups of that factor (e.g., Method A, B, C).
- **F-value:** compares how much groups differ from each other vs. within themselves.

When and Why We Use It?

- Use it when you have **one factor with 3 or more groups**, and the **data is numeric**.
- **Doing many t-tests** instead **raises the chance of a wrong result** (Type-I error).
- **One-Way ANOVA checks all group means together in a single test.**

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Steps to Calculate using One-Way ANOVA

$$F = \frac{\text{Mean Square Between Groups (MSB)}}{\text{Mean Square Within Groups (MSW)}}$$

- $MSB = SSB \div (k-1)$; $MSW = SSW \div (N-k)$.
- **A big F value (with $p < 0.05$) means at least one group mean is different.**

Step 1: Write H_0 (all means equal) and H_1 (at least one is different).

Step 2: Find the mean of each group and the overall (grand) mean.

Step 3: Find SSB (Sum of Squares Between groups) and SSW (Sum of Squares Within groups).

Step 4: Find the degrees of freedom (df): $df(\text{between}) = k-1$, $df(\text{within}) = N-k$.

Step 5: Find the Mean Squares: $MSB = SSB \div df(\text{between})$, $MSW = SSW \div df(\text{within})$.

Step 6: Find $F = MSB \div MSW$.

Step 7: Compare F with the table (critical) value, and give your answer.

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One-Way ANOVA: Problem

For Example:

A researcher wants to check if three fertilizers give different average crop yield (kg per plot). Five plots were tested for each fertilizer.

Fertilizer A	Fertilizer B	Fertilizer C
51	23	56
45	43	76
33	23	74
45	43	87
67	45	56

Using $\alpha = 0.05$, check if the average yield is the same for all three fertilizers.
(Given: table value of F at $df = 2, 12$ and $\alpha = 0.05$ is 3.89.)

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One-Way ANOVA: Setting Up

- H_0 :** $\mu_1 = \mu_2 = \mu_3$ (all three fertilizers give the same average yield) **H_a :** at least one is different.
- Here, $k = 3$ groups, $n = 5$ observations per group, so $N = k \times n = 15$.
- Step 1** — Add up the values in each group to get the group totals:
 - $T_A = 51 + 45 + 33 + 45 + 67 = 241$
 - $T_B = 23 + 43 + 23 + 43 + 45 = 177$
 - $T_C = 56 + 76 + 74 + 87 + 56 = 349$
- Step 2** — Divide each group total by n to get the group means:
 - Mean of A** = $241 \div 5 = 48.2$
 - Mean of B** = $177 \div 5 = 35.4$
 - Mean of C** = $349 \div 5 = 69.8$
- Step 3** — Add the group totals to get the grand total, then divide by N for the grand mean:
 - Grand total T** = $241 + 177 + 349 = 767$
 - Grand mean** = $767 \div 15 = 51.13$

- k = number of groups.
- n = number of observations in each group.
- N = total number of observations ($k \times n$).
- T = grand total (sum of every value in the data).

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One-Way ANOVA: Correction Factor and Total SS

- Correction Factor (CF)** — removes the effect of the overall average before we measure spread. $CF = T^2 \div N$.
- Step 4** — Calculate the Correction Factor:
 - CF** = $767^2 \div 15 = 588289 \div 15 = 39219.27$
- Step 5** — Square every single value in the data and add them all up (ΣX^2):
 - Group A:** $51^2 + 45^2 + 33^2 + 45^2 + 67^2 = 12229$
 - Group B:** $23^2 + 43^2 + 23^2 + 43^2 + 45^2 = 6781$
 - Group C:** $56^2 + 76^2 + 74^2 + 87^2 + 56^2 = 25093$
 - ΣX^2** = $12229 + 6781 + 25093 = 44103$
- Step 6** — Subtract CF from ΣX^2 to get the Total Sum of Squares (SST). This measures the **total spread of every value around the grand mean**:
 - SST** = $\Sigma X^2 - CF = 44103 - 39219.27 = 4883.73$

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One-Way ANOVA: Between-Groups Sum of Squares

- SSB measures **how much the group means differ from the grand mean** — this is the variation "between" groups.
- $SSB = (T_A^2 \div n) + (T_B^2 \div n) + (T_C^2 \div n) - CF$
- Step 7** — Square each group total and divide by n :
 - $T_A^2 \div n = 241^2 \div 5 = 58081 \div 5 = 11616.20$
 - $T_B^2 \div n = 177^2 \div 5 = 31329 \div 5 = 6265.80$
 - $T_C^2 \div n = 349^2 \div 5 = 121801 \div 5 = 24360.20$
- Step 8** — Add these three values together:
 - Sum** = $11616.20 + 6265.80 + 24360.20 = 42242.20$
- Step 9** — Subtract CF to get SSB:
 - SSB** = $42242.20 - 39219.27 = 3022.93$

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One-Way ANOVA: SSW, df, and Mean Squares

- SSW measures **how much the individual values differ from their own group mean** — the variation "within" groups.
- Step 10** — Since $SST = SSB + SSW$, we can find SSW by subtraction:
 $SSW = SST - SSB = 4883.73 - 3022.93 = 1860.80$
- Step 11** — Find the **degrees of freedom (df)**, which count **how many independent comparisons we have**:
 - $df(\text{between}) = k - 1 = 3 - 1 = 2$
 - $df(\text{within}) = N - k = 15 - 3 = 12$
 - $df(\text{total}) = N - 1 = 14$ (check: $2 + 12 = 14$ ✓)
- Step 12** — Divide each SS by its own df to get the Mean Squares (the "average" variation per df):
 - $MSB = SSB \div df(\text{between}) = 3022.93 \div 2 = 1511.47$
 - $MSW = SSW \div df(\text{within}) = 1860.80 \div 12 = 155.07$

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One-Way ANOVA: Final Table and Conclusion

- Step 13** — Find $F = MSB \div MSW = 1511.47 \div 155.07 = 9.75$

Source of Variation	SS	df	MS	F
Between Groups	3022.93	2	1511.47	9.75
Within Groups	1860.80	12	155.07	
Total	4883.73	14		

- Step 14** — Compare with the table value:

$F(2, 12)$ at $\alpha = 0.05$ is **3.89**. Since our **calculated $F = 9.75$ is bigger than 3.89**, we **reject H_0** . The three fertilizers really do give significantly different average yields.

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State the procedure and example to illustrate the completely randomized design in a two-way ANOVA?

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Two-Way ANOVA

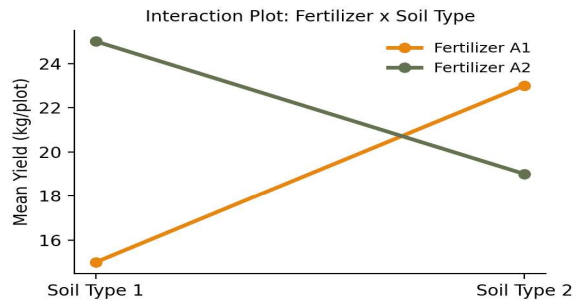
Two-Way ANOVA checks the effect of **two factors** on a numeric outcome at the same time. It also checks whether the two factors affect the result together (this is called **interaction**).

Key Concepts of Two-Way ANOVA are:

- Main effect**: how much one factor changes the result, on its own.
- Interaction effect**: when one factor's effect changes depending on the other factor.
- Two-Way ANOVA tests both main effects and the interaction, all at once.**

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What Does Interaction Look Like?



The two lines cross each other. This tells us that fertilizer A1 gives a low yield in Soil Type 1 but a high yield in Soil Type 2, while fertilizer A2 shows the opposite pattern — the effect of fertilizer clearly depends on soil type. This is called interaction.

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When to Use Two-Way ANOVA

When and Why We Two-Way ANOVA?:

- Use it when the data is grouped by **two different factors at once**.
- It shows the effect of **each factor separately**, and also whether **they interact**.
- **Example:** checking fertilizer type and soil type together on crop yield.
- **Two-Way ANOVA gives more complete information than running two separate One-Way ANOVA tests.**

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Steps to Calculate Two-Way ANOVA

$$F(A) = MS(A) \div MSE$$

$$F(B) = MS(B) \div MSE$$

$$F(A \times B) = MS(A \times B) \div MSE$$

- MSE (Mean Square Error) comes from the variation left over within each cell.
- A big F value (with $p < 0.05$) means that factor, or the interaction, has a real effect.

Step 1: Write H_0 for Factor A, Factor B, and their interaction (each says "no effect").

Step 2: Put the data in a two-way table; find the mean of each row, each column, and the overall mean.

Step 3: Split the total variation (SST) into 4 parts — SSA, SSB, $SS(A \times B)$, and SSE — using the row, column, and overall means.

Step 4: Find the degrees of freedom for each part: $df(A) = a - 1$, $df(B) = b - 1$, $df(A \times B) = (a - 1)(b - 1)$, $df(error) = ab(n - 1)$.

Step 5: Divide each SS by its own df to get the Mean Squares — MSA, MSB, $MS(A \times B)$, MSE.

Step 6: Find the three F-values by dividing each Mean Square by MSE.

Step 7: Compare each F-value with its table (critical) value, and decide if that effect is significant.

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Two-Way ANOVA: Problem

For Example:

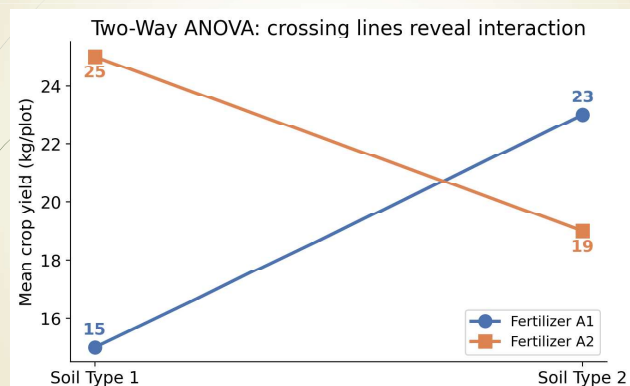
An experiment checks two fertilizers (Factor A) and two soil types (Factor B) on crop yield (kg/plot). Each combination was tested twice.

	Soil Type 1	Soil Type 2
Fertilizer A1	14, 16	22, 24
Fertilizer A2	26, 24	20, 18

Using $\alpha = 0.05$, check if fertilizer type, soil type, or both together affect yield.

(Given: table value of F at $df = 1, 4$ and $\alpha = 0.05$ is 7.71.)

Visualizing the Two-Way ANOVA Interaction



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Two-Way ANOVA: Setting Up

- H₀(A):** both fertilizers give the same mean yield. **H₀(B):** both soil types give the same mean yield. **H₀(A×B):** no interaction.
- Here, $a = 2$ (fertilizers), $b = 2$ (soil types), $n = 2$ (replications), so $N = a \times b \times n = 8$.
- Step 1** — Add up the replications in each cell to get the cell totals:
 - $T_{A1S1} = 14 + 16 = 30$
 - $T_{A1S2} = 22 + 24 = 46$
 - $T_{A2S1} = 26 + 24 = 50$
 - $T_{A2S2} = 20 + 18 = 38$
- Step 2** — Add across each row and each column:
 - Row totals:** $T_{A1} = 30 + 46 = 76$ $T_{A2} = 50 + 38 = 88$
 - Column totals:** $T_{S1} = 30 + 50 = 80$ $T_{S2} = 46 + 38 = 84$

- a, b = number of levels of Factor A and Factor B.
- n = number of replications in each cell.
- $N = a \times b \times n$ = total number of observations.

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Two-Way ANOVA: Correction Factor and Total SS

- Step 3** — Add the row (or column) totals to get the grand total, then divide by N :
 - Grand total $T = 76 + 88 = 164$** (check: $80 + 84 = 164$ ✓)
 - Grand mean $= 164 \div 8 = 20.5$**
- Step 4** — Calculate the Correction Factor, $CF = T^2 \div N$:
 - $CF = 164^2 \div 8 = 26896 \div 8 = 3362$**
- Step 5** — Square every one of the 8 raw values and add them up (ΣX^2):
 - $\Sigma X^2 = 14^2 + 16^2 + 22^2 + 24^2 + 26^2 + 24^2 + 20^2 + 18^2 = 3488$**
- Step 6** — Subtract CF from ΣX^2 to get the Total Sum of Squares (SST):
 - $SST = \Sigma X^2 - CF = 3488 - 3362 = 126$**

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Two-Way ANOVA: Sum of Squares for A and B

- Step 7** — Find SSA (Factor A). Square each row total, divide by $(b \times n)$, add, then subtract CF:
 - $T_{A1}^2 \div (b \times n) = 76^2 \div 4 = 5776 \div 4 = 1444$
 - $T_{A2}^2 \div (b \times n) = 88^2 \div 4 = 7744 \div 4 = 1936$
 - $SSA = (1444 + 1936) - CF = 3380 - 3362 = 18$**
- Step 8** — Find SSB (Factor B). Square each column total, divide by $(a \times n)$, add, then subtract CF:
 - $T_{S1}^2 \div (a \times n) = 80^2 \div 4 = 6400 \div 4 = 1600$
 - $T_{S2}^2 \div (a \times n) = 84^2 \div 4 = 7056 \div 4 = 1764$
 - $SSB = (1600 + 1764) - CF = 3364 - 3362 = 2$**

25 Two-Way ANOVA: Interaction SS and Error SS

- **Step 9** — First find $SS(\text{cells})$: square each cell total, divide by n , and add them up:
 - **Sum** = $30^2 \div 2 + 46^2 \div 2 + 50^2 \div 2 + 38^2 \div 2 = 450 + 1058 + 1250 + 722 = 3480$
 - **$SS(\text{cells})$** = $3480 - CF = 3480 - 3362 = 118$
- **Step 10** — The interaction SS is whatever is left in $SS(\text{cells})$ after removing SSA and SSB :
 - **$SS(A \times B)$** = $SS(\text{cells}) - SSA - SSB = 118 - 18 - 2 = 98$
- **Step 11** — The error SS is whatever is left in SST after removing $SS(\text{cells})$ — this is the variation within each cell (between repeated trials):
 - **SSE** = $SST - SS(\text{cells}) = 126 - 118 = 8$

26 Two-Way ANOVA: df and Mean Squares

- **Step 12** — Find the degrees of freedom for each part:
 - **$df(A) = a - 1 = 2 - 1 = 1$**
 - **$df(B) = b - 1 = 2 - 1 = 1$**
 - **$df(A \times B) = (a - 1)(b - 1) = 1 \times 1 = 1$**
 - **$df(\text{error}) = ab(n - 1) = 2 \times 2 \times 1 = 4$** (check: $1 + 1 + 1 + 4 = 7 = N - 1$ ✓)
- **Step 13** — Divide each SS by its own df to get the Mean Squares:
 - **$MSA = 18 \div 1 = 18.0$**
 - **$MSB = 2 \div 1 = 2.0$**
 - **$MS(A \times B) = 98 \div 1 = 98.0$**
 - **$MSE = 8 \div 4 = 2.0$**

27 Two-Way ANOVA: Final Table and Conclusion

- **Step 14** — Find each F-value by dividing its Mean Square by $MSE (=2.0)$:
 - **$F(A) = 18.0 \div 2.0 = 9.0$** **$F(B) = 2.0 \div 2.0 = 1.0$** **$F(A \times B) = 98.0 \div 2.0 = 49.0$**
- | Source | SS | df | MS | F |
|-------------------|----|----|------|------|
| Fertilizer (A) | 18 | 1 | 18.0 | 9.0 |
| Soil Type (B) | 2 | 1 | 2.0 | 1.0 |
| Interaction (A×B) | 98 | 1 | 98.0 | 49.0 |
| Error | 8 | 4 | 2.0 | |
- **Step 15** — Compare with the table values: $F(1,4)$ at $\alpha = 0.05$ is 7.71. Fertilizer ($F=9.0$) is significant, soil type ($F=1.0$) is not, but the interaction ($F=49.0$) is highly significant. This matches the crossing lines we saw earlier — **the effect of fertilizer really does depend on soil type.**

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What is a factorial design?

29 Factorial Design: Definition and Key Concepts

A **factorial design** is an experiment in which we study **two or more factors at the same time**, testing every possible combination of their levels. This lets us measure each factor's own effect and their **interaction**, together in **one experiment**.

Key Concepts of Factorial Design are:

- **Treatment combination**: one specific pairing of levels, one from each factor (e.g., Fertilizer A1 with Soil S1).
- A **2×2 factorial design** means 2 factors, each with 2 levels, giving $2 \times 2 = 4$ treatment combinations.
- A **Two-Way ANOVA (which we have already covered)** is exactly how a factorial design is analysed.

30 Factorial Design: Example Layout

- The fertilizer-and-soil-type experiment solved earlier is a **2×2 factorial design**:

2 fertilizers × 2 soil types = 4 treatment combinations, each tested twice.

	Soil Type 1	Soil Type 2
Fertilizer A1	Treatment A1S1	Treatment A1S2
Fertilizer A2	Treatment A2S1	Treatment A2S2

Each box is a treatment combination. We already calculated the main effect of fertilizer, the main effect of soil type, and their interaction using Two-Way ANOVA — that worked example is a complete illustration of analysing a factorial design.

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Define Latin square design.

32 Latin Square Design: Definition and Key Concepts

A Latin Square design is used when we want to control **two nuisance factors** (row factor and column factor) while studying **one treatment factor**. Each treatment appears exactly once in every row and exactly once in every column.

Key Concepts of Latin Square Design are:

- An **n×n Latin Square** has n rows, n columns, and n treatments (shown as letters A, B, C, ...).
- It needs far fewer iterations than testing every row×column×treatment combination separately.
- Used in agriculture (e.g., field trials), and in industry when two background factors (like machine and operator) might affect the result.

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Steps to Calculate a Latin Square ANOVA

- **SST** = SS(rows) + SS(columns) + SS(treatments) + SSE
- For an $n \times n$ square: **$N = n^2$ observations** in total.

Step 1: Write H_0 for rows, columns, and treatments (each says "no effect").

Step 2: Find the Correction Factor: $CF = T^2 \div N$, where T is the grand total.

Step 3: Find the Total SS: $SST = \sum X^2 - CF$ (square every single value and add them up).

Step 4: Find SS(rows), SS(columns), and SS(treatments), each as (sum of squared totals $\div n$) - CF.

Step 5: Find SSE by subtraction: $SSE = SST - SS(\text{rows}) - SS(\text{columns}) - SS(\text{treatments})$.

Step 6: Find df: rows = $n-1$, columns = $n-1$, treatments = $n-1$, error = $(n-1)(n-2)$.

Step 7: Find Mean Squares and F-values, then compare with the table value to decide significance.

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Latin Square: Problem

For Example:

Three drivers (rows) each test-drive three cars (columns), each fitted with a different tyre brand A, B, or C (the treatment), arranged so each brand is used once per driver and once per car. Fuel efficiency (km/l) is recorded.

	Car 1	Car 2	Car 3
Driver 1	A = 14	B = 18	C = 20
Driver 2	B = 16	C = 22	A = 12
Driver 3	C = 19	A = 13	B = 17

Using $\alpha = 0.05$, test whether tyre brand (treatment) affects fuel efficiency.

(Given: table value of F at $df = 2, 2$ and $\alpha = 0.05$ is 19.0.)

Visualizing the Latin Square Layout

Latin Square: each tyre brand appears once per driver, once per car

Driver	Car 1	Car 2	Car 3
Driver 1	A=14	B=18	C=20
Driver 2	B=16	C=22	A=12
Driver 3	C=19	A=13	B=17

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Latin Square

- **Step 1** — Row totals: Driver1 = $14+18+20 = 52$; Driver2 = $16+22+12 = 50$; Driver3 = $19+13+17 = 49$
- **Step 2** — Column totals: Car1 = $14+16+19 = 49$; Car2 = $18+22+13 = 53$; Car3 = $20+12+17 = 49$
- **Step 3** — Treatment totals: A = $14+12+13 = 39$; B = $18+16+17 = 51$; C = $20+22+19 = 61$
- **Grand total T** = $52+50+49 = 151$ (check: $49+53+49 = 151 \checkmark$, and $39+51+61 = 151 \checkmark$)
- **Correction Factor CF** = $151^2 \div 9 = 22801 \div 9 = 2533.44$
- **($\sum X^2$)** = $14^2+18^2+20^2+16^2+22^2+12^2+19^2+13^2+17^2 = 2623$
- **Total SS** = $\sum X^2 - CF = 2623 - 2533.44 = 89.56$
- **SS(rows)** = $(52^2+50^2+49^2) \div 3 - CF = 2535.00 - 2533.44 = 1.56$
- **SS(columns)** = $(49^2+53^2+49^2) \div 3 - CF = 2537.00 - 2533.44 = 3.56$
- **SS(treatments)** = $(39^2+51^2+61^2) \div 3 - CF = 2614.33 - 2533.44 = 80.89$
- **SSE** = Total SS - SS(rows) - SS(columns) - SS(treatments) = $89.56 - 1.56 - 3.56 - 80.89 = 3.55$

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Latin Square: Final Table and Conclusion

- df: rows = columns = treatments = $n-1 = 2$; error = $(n-1)(n-2) = 2 \times 1 = 2$

Source	SS	df	MS	F
Rows (Driver)	1.56	2	0.78	0.44
Columns (Car)	3.56	2	1.78	1.00
Treatments (Tyre)	80.89	2	40.44	22.75
Error	3.55	2	1.78	

Table value of $F(2,2)$ at $\alpha = 0.05$ is 19.0. Since $F(\text{treatments}) = 22.75 > 19.0$, we reject H_0 — tyre brand significantly affects fuel efficiency. Driver ($F=0.44$) and car ($F=1.00$) are not significant, showing the design successfully controlled these two nuisance factors.

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Class-16: Non-parametric Tests

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Learning Objective Class 16

By the end of the class, a student should be able to answer the following Questions

- Define and differentiate between the parametric and non-parametric tests?
- What is a chi-square test and illustrate a chi-square test for (a) the goodness of fit, (b) independence of variables, (c) the equality of more than two population proportions?
- Explain the run test of randomness for metric and non-metric data
- Describe the one-sample and two-sample sign tests
- What is Mann-Whitney U test?
- What is Wilcoxon signed rank tests?
- What is Kruskal-Wallis tests?

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Define and differentiate between the parametric and non-parametric tests?

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Parametric vs. Non-Parametric Tests

Aspect	Parametric Tests	Non-Parametric Tests
Population assumption	Assumes a specific distribution (usually normal)	Makes little or no assumption about the distribution
Data needed	Interval or ratio (numeric) data	Works with ordinal, nominal, or ranked data
Power	More powerful, if its assumptions hold	Less powerful, but safer when assumptions fail or samples are small
Examples	t-test, Z-test, One-Way / Two-Way ANOVA	Chi-square, Sign test, Wilcoxon, Kruskal-Wallis, Run test

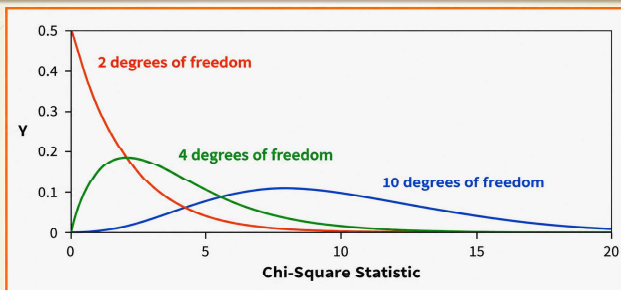
Rule of thumb: use a parametric test when the data is numeric and roughly normal; switch to a non-parametric test when the data is ranked/categorical, the sample is small, or the normality assumption is doubtful.

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What is a chi-square test and illustrate a chi-square test for (a) the goodness of fit, (b) independence of variables, (c) the equality of more than two population proportions?

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Chi-Square Test: Definition



- The **Chi-Square (χ^2) test** checks if the **numbers we observe in categories** are **different from the numbers we expect**. It is **used for categorical data**, like **yes/no** or **male/female**, **not for numeric data**.
- The Chi-Square curve is **not symmetric**, and its **shape depends on the degrees of freedom**.

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Types of Chi-Square Tests and its Use

Type	Purpose
(a) Goodness-of-Fit Test	Checks if observed frequencies of ONE categorical variable match an expected pattern
(b) Test for Independence	Checks if TWO categorical variables are related to each other
(c) Test for Equality of Proportions	Checks if a proportion is the same across THREE or more populations

When and Why to use Chi-Square Test?:

- Use it for **categorical data (counts)**, where **we cannot find a mean**.
- Each category should ideally have an **expected count of 5 or more**.
- Common examples:** gender, preference, pass/fail, product quality, yes/no.
- It lets us test relationships or patterns in survey and count data without needing numeric measurements.**

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Steps to Calculate the Chi-Square Test

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- **O** = the count we actually observe; **E** = the count we expect.
- This same procedure applies to all three types (a), (b), and (c).

Step 1: Write H_0 (no difference / no relationship) and H_1 (there is a difference / relationship).

Step 2: Arrange the observed counts (O) in a table, with row totals, column totals, and the grand total.

Step 3: Find the expected count (E) for each box or category.

Step 4: For every box, calculate $(O - E)^2 \div E$.

Step 5: Add up all the $(O - E)^2 \div E$ values to get the Chi-Square value (χ^2).

Step 6: Find the degrees of freedom (df) — the formula depends on the type of test.

Step 7: Compare χ^2 with the table (critical) value, and decide whether to reject H_0 .

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Chi-Square (a) Goodness of Fit: Problem

For Example:

A die is rolled 120 times to check whether it is fair. The number of times each face turned up is recorded below.

Face	1	2	3	4	5	6
Observed	22	17	20	26	18	17

H_0 : the die is fair (every face has an equal chance).

Test at $\alpha = 0.05$.

(Given: table value of χ^2 at $df = 5$ and $\alpha = 0.05$ is 11.07.)

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Chi-Square (a) Goodness of Fit: Solution

- If the die is fair, every face is equally likely, so each face's expected count is the total divided by 6 (k = number of categories = 6):

■ **Expected count E** = $120 \div 6 = 20$ for every face

■ **Degrees of freedom: df** = $k - 1 = 6 - 1 = 5$

Face	O	E	O-E	(O-E) ²	(O-E) ² /E
1	22	20	2	4	0.20
2	17	20	-3	9	0.45
3	20	20	0	0	0.00
4	26	20	6	36	1.80
5	18	20	-2	4	0.20
6	17	20	-3	9	0.45

$$\chi^2 = 0.20 + 0.45 + 0.00 + 1.80 + 0.20 + 0.45 = 3.10.$$

Table value of $\chi^2(df=5)$ at $\alpha = 0.05$ is 11.07. **Since 3.10 is smaller than 11.07, we fail to reject H_0 — the die appears to be fair.**

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Chi-Square (b) Independence: Problem

For Example:

A survey asks men and women how they travel to work — by bus or by car.

	Bus	Car	Row Total
Men	30	70	100
Women	50	50	100
Column Total	80	120	200

H_0 : gender and travel choice are not related. Test at $\alpha = 0.05$.

(Given: table value of χ^2 at $df = 1$ and $\alpha = 0.05$ is 3.84.)

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Chi-Square (b) Independence: Expected Frequencies

• **Expected frequency (E)** — the count we would expect in a cell if H_0 (independence) were true.

• Formula: $E = (\text{row total} \times \text{column total}) \div \text{grand total}$.

• **Steps** — Recap the totals from the table: Row totals: Men = 100, Women = 100. Column totals: Bus = 80, Car = 120. Grand total: $N = 200$.

- $E(\text{Men, Bus}) = (100 \times 80) \div 200 = 40$
- $E(\text{Men, Car}) = (100 \times 120) \div 200 = 60$
- $E(\text{Women, Bus}) = (100 \times 80) \div 200 = 40$
- $E(\text{Women, Car}) = (100 \times 120) \div 200 = 60$
- **Degrees of freedom: $df = (2-1) \times (2-1) = 1$**

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Chi-Square (b) Independence: Computation and Conclusion

Group	O	E	O-E	(O-E) ²	(O-E) ² /E
Men, Bus	30	40	-10	100	2.50
Men, Car	70	60	10	100	1.67
Women, Bus	50	40	10	100	2.50
Women, Car	50	60	-10	100	1.67

$$\chi^2 = 2.50 + 1.67 + 2.50 + 1.67 = 8.34.$$

Table value of χ^2 ($df=1$) at $\alpha = 0.05$ is 3.84. **Since 8.34 is bigger than 3.84, we reject H_0 — gender and travel choice are significantly related.**

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Chi-Square (c) Equality of Proportions: Problem

For Example:

Three machines produce items. A sample of 100 items was checked from each machine for defects.

	Machine A	Machine B	Machine C
Defective	10	15	20
Non-Defective	90	85	80

H_0 : the proportion of defective items is the same for all three machines. Test at $\alpha = 0.05$.
(Given: table value of χ^2 at $df = 2$ and $\alpha = 0.05$ is 5.99.)

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Chi-Square (c) Equality of Proportions: Expected Frequencies

- **Calculate Row totals:** Defective = $10+15+20 = 45$; Non-Defective = $90+85+80 = 255$.
- **Calculate Column totals:** Machine A = 100, Machine B = 100, Machine C = 100; Grand total $N = 300$.
 - $E(\text{Defective, A}) = (45 \times 100) \div 300 = 15$
 - $E(\text{Defective, B}) = (45 \times 100) \div 300 = 15$
 - $E(\text{Defective, C}) = (45 \times 100) \div 300 = 15$
 - $E(\text{Non-Defective, A}) = (255 \times 100) \div 300 = 85$
 - $E(\text{Non-Defective, B}) = (255 \times 100) \div 300 = 85$
 - $E(\text{Non-Defective, C}) = (255 \times 100) \div 300 = 85$
- **Degrees of freedom: $df = (2-1) \times (3-1) = 2$**

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Chi-Square (χ^2) Equality of Proportions: Computation and Conclusion

Cell	O	E	O-E	(O-E) ²	(O-E) ² /E
Defective, A	10	15	-5	25	1.67
Defective, B	15	15	0	0	0.00
Defective, C	20	15	5	25	1.67
Non-Def., A	90	85	5	25	0.29
Non-Def., B	85	85	0	0	0.00
Non-Def., C	80	85	-5	25	0.29

$\chi^2 = 1.67 + 0.00 + 1.67 + 0.29 + 0.00 + 0.29 = 3.92$. Table value of χ^2 (df=2) at $\alpha = 0.05$ is 5.99.

Since 3.92 is smaller than 5.99, we fail to reject H_0 — the defect proportions are not significantly different across the three machines.

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Explain the run test of randomness for metric and non-metric data

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Run Test: Definition and Key Concepts

The **run test** also called as **Wald-Wolfowitz test**, checks whether a **sequence of observations occurred by chance**, i.e. whether the order is random, rather than following some pattern.

Key Concepts of the Run Test are:

- **Run**: an unbroken sequence of the same symbol (e.g., MMM is one run of 3).
- **Non-metric data** (already categories, like M/F) can be used directly.
- **Metric data** (numeric) is first converted: values above the median are marked A, values below are marked B.

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Run Test: Formula and Steps

- n_1, n_2 = counts of the two symbols; R = number of runs observed.
- Z follows (approximately) a standard normal distribution for reasonably large n_1, n_2 .

Step 1: Write H_0 : the sequence is random, H_1 : the sequence is not random.

Step 2: If the data is metric, convert each value to A (above median) or B (below median).

Step 3: Count n_1, n_2 (how many of each symbol) and R (the number of runs).

Step 4: Find the expected number of runs: $\mu R = (2n_1n_2)/(n_1+n_2) + 1$.

Step 5: Find the variance: $\sigma^2 R = [2n_1n_2(2n_1n_2 - n_1 - n_2)] / [(n_1+n_2)^2(n_1+n_2-1)]$.

Step 6: Find $Z = (R - \mu R) / \sigma R$.

Step 7: Compare $|Z|$ with the table value (1.96 at $\alpha=0.05$), and decide whether to reject H_0 .

Run Test: Problem

For Example:

Healthy (H) and diseased (D) mango trees were planted along a road, in this order of occurrence:

HH | DD | HHHHH | DDD | HHHH | DDDDD | HHHHHHHHH

Marking off the runs (underlined groups) gives 7 runs:

$n_1 = 20$ (H), $n_2 = 10$ (D), $R = 7$.

H_0 : the trees were planted in random order.

Test at $\alpha = 0.01$.

(Given: critical Z at $\alpha = 0.01$, two-tailed, is ± 2.58 .)

Source: Kothari, C.R., *Research Methodology: Methods and Techniques*, New Age International Publishers, p. 300.

Visualising the Run Test Problem

Run Test: 7 runs in the Healthy(H) / Diseased(D) tree sequence



$n_1 = 20$ (H), $n_2 = 10$ (D), $R = 7$ runs observed

Run Test: Solution and Conclusion

- Step - 1: Calculate Expected runs $\mu R = (2 \times 20 \times 10) / (20 + 10) + 1 = 400 / 30 + 1 = 13.33 + 1 = 14.33$
- Step - 2: Calculate the Variance $\sigma^2 R = [2 \times 20 \times 10 \times (400 - 30)] \div [30^2 \times 29] = (400 \times 370) \div 26100 = 148000 \div 26100 = 5.67$
- Step - 3: Calculate $\sigma R = \sqrt{5.67} = 2.38$

Using $z = 2.58$ ($\alpha = 0.01$, two-tailed), the acceptance region is $\mu R \pm 2.58\sigma R = 14.33 \pm 6.14$, i.e. from 8.19 to 20.47.

The observed $R = 7$ falls outside this region (below 8.19), so we reject H_0 — there is a strong indication that the diseased trees come in a non-random grouping, not a random scatter.

Describe the one-sample and two-sample sign tests

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Sign Test: Definition and Key Concepts

The **sign test** is a simple non-parametric test that only looks at the **direction (sign)** of a difference, ignoring its size.

It is **used when data cannot be assumed normal**.

Key Concepts of the Sign Test are:

- **One-sample sign test:** compares each value with a hypothesised median (each is marked + or -).
- **Two-sample (paired) sign test:** compares two related values (e.g., before/after) and marks the sign of their difference.
- **Values exactly equal to the median (or paired differences of 0) are dropped from the count.**

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Sign Test: Steps to Calculate

Step 1: Write H_0 (no difference from the hypothesised median, or no difference between the pair).

Step 2: For each observation (or pair), find the sign: + if above the median (or if it increased), - if below (or decreased). Drop any exact ties (0's).

Step 3: Count the number of + signs and the number of - signs; let n = total signs (after dropping ties).

Step 4: Take the smaller of the two counts as the test statistic.

Step 5: Compare this smaller count with the critical value from the sign-test table for that n and α .

Step 6: If the smaller count is less than or equal to the table value, reject H_0 .

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One-Sample Sign Test: Problem and Solution

For Example:

Playing four rounds of golf, 11 professionals totalled:

280, 282, 290, 273, 283, 283, 275, 284, 282, 279, 281.

Test H_0 : median = 284 against H_a : median < 284.

Solution:

• Compare each score with 284:

• -, -, +, -, -, -, -, (tie(dropped)), -, -, -

After dropping the tie (one score = 284), $n = 10$; Number of + = 1; Number of - = 9

The tie is removed before the probability calculation.

• Using Binomial Distribution, calculate the probability:

$P(1 \text{ or fewer } + \text{ out of } 10, p=0.5) = {}^{10}C_1(0.5)^1(0.5)^9 + {}^{10}C_0(0.5)^0(0.5)^{10} = 0.010 + 0.001 = 0.011$

Since 0.011 is less than $\alpha = 0.05$, we reject H_0 — professional golfers' average is significantly less than 284 for four rounds.

Source: Kothari, C.R., Research Methodology: Methods and Techniques, New Age International Publishers, p. 285.

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Two-Sample Sign Test: Problem and Solution

For Example:

Two archaeologists, X and Y, each counted the artifacts dug up at 30 common sites. Comparing X's count with Y's count at each site gave 20 sites where $X > Y$ (+), 6 sites where $X < Y$ (-), and 4 tied sites (dropped). Test at $\alpha = 0.01$.

(Given: critical Z for a one-tailed test at $\alpha = 0.01$ is 2.32.)

Solution:

• After dropping ties, $n = 26$; observed proportion of + = $20 \div 26 = 0.7692$

• Standard error ($H_0: p=0.5$) = $\sqrt{(0.5 \times 0.5 \div 26)} = 0.0981$

• Critical proportion (one-tailed, $z=2.32$ at $\alpha=0.01$) = $0.5 + 2.32 \times 0.0981 = 0.7276$

Consider H_0 : the two archaeologists are equally good; and H_a : X is better.

Since the observed proportion 0.7692 exceeds the critical value 0.7276, it falls in the rejection region — we reject H_0 and conclude archaeologist X is better at finding artifacts.

Source: Kothari, C.R., Research Methodology: Methods and Techniques, New Age International Publishers, p. 286.

What is Wilcoxon signed rank tests?

Wilcoxon Signed Rank Test: Definition and Key Concepts

The **Wilcoxon signed rank test** is a non-parametric test for **paired data** (like before/after). Unlike the sign test, it uses both the **direction and the size** of each difference, by ranking them — making it more powerful than the sign test.

Key Concepts of the Wilcoxon Signed Rank Test are:

- **W^+ , W^-** : the sum of the ranks of the positive differences, and of the negative differences.
- **Tied absolute differences** are given the average of the ranks they would have occupied.

Wilcoxon Test: Steps to Calculate

Step 1: Write H_0 (no difference between the pair of measurements).

Step 2: Find the difference (After – Before) for every pair; drop any pair with a difference of 0.

Step 3: Take the absolute value of every difference, then rank them from smallest (1) to largest; give tied values the average rank.

Step 4: Reattach the original sign (+ or –) to each rank.

Step 5: Add up all the positive ranks to get W^+ , and all the negative ranks to get W^- .

Step 6: Take W = the smaller of W^+ and W^- as the test statistic.

Step 7: Compare W with the critical value from the Wilcoxon table; reject H_0 if W is less than or equal to it.

Wilcoxon Test: Problem

For Example

An experiment on the effect of brand name on quality perception: 16 subjects tasted and scored two samples, Brand A and Brand B, on an ordinal scale.

Pair	1	2	3	4	5	6	7	8
Brand A	73	43	47	53	58	47	52	58
Brand B	51	41	43	41	47	32	24	58

Pair	9	10	11	12	13	14	15	16
Brand A	38	61	56	56	34	55	65	75
Brand B	43	53	52	57	44	57	40	68

H_0 : no difference between the perceived quality of the two samples. Test at $\alpha = 0.05$.

(Given: after dropping any tied pair, the table value of T at $\alpha = 0.05$, two-tailed, is 25 for $n = 15$.)

Source: Kothari, C.R., *Research Methodology: Methods and Techniques*, New Age International Publishers, p. 292.

Wilcoxon Test: Solution

- **Steps :** Find each difference (A-B), drop ties, then rank the absolute differences from smallest to largest (averaging tied ranks), and reattach the original sign:
 - **Positive differences (pairs 1,2,3,4,5,6,7,10,11,15,16), ranks sum to $W^+ = 101.5$**
 - **Negative differences (pairs 9,12,13,14), ranks sum to $W^- = 18.5$** (pair 8 is dropped: A=B=58, difference = 0)
 - **Check: $W^+ + W^- = 101.5 + 18.5 = 120 = n(n+1) \div 2 = 15 \times 16 \div 2 \checkmark$** (n = 15 after dropping 1 tie)
 - **Test statistic T = smaller of the two = 18.5**

Table value of T for n = 15 at $\alpha = 0.05$ (two-tailed) is 25.

Since $T = 18.5$ is less than 25, we reject H_0 — there is a significant difference between the perceived quality of the two brand samples.

What is Kruskal-Wallis tests?

Kruskal-Wallis Test: Definition and Key Concepts

The **Kruskal-Wallis test** is the **non-parametric version of One-Way ANOVA**. It compares 3 or more groups, but uses the **ranks** of the data instead of the raw values, so it does not need a normal distribution.

Key Concepts of the Kruskal-Wallis Test are:

- **H statistic:** the test statistic, which approximately follows a Chi-Square distribution.
- **R_i :** the sum of ranks for group i ;
- **n_i :** the number of observations in group i ;
- **N :** total observations.

Kruskal-Wallis Test: Formula and Steps

- $H = [12 \div (N(N+1))] \times \sum (R_i^2 \div n_i) - 3(N+1)$
- $df = k - 1$, where k = number of groups. Compare H with the Chi-Square table.

Step 1: Write H_0 : all groups have the same distribution (equal medians).

Step 2: Combine all the data from every group into one list, and rank them from smallest (1) to largest; average the ranks for ties.

Step 3: Add up the ranks that belong to each group to get R_i for each group.

Step 4: Calculate H using the formula above.

Step 5: Find $df = k - 1$, and compare H with the Chi-Square table value; reject H_0 if H is bigger.

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Kruskal-Wallis: Problem

For Example:

A bowler tests four bowling balls (A, B, C, D), bowling five games with each.

Test H_0 : the bowler performs equally well with all four balls.

Ball A	Ball B	Ball C	Ball D
271	252	260	279
282	275	255	242
257	302	239	297
248	268	246	270
262	276	266	258

Test at $\alpha = 0.05$.

(Given: table value of χ^2 at $df = 3$ and $\alpha = 0.05$ is 7.815.)

Source: Kothari, C.R., *Research Methodology: Methods and Techniques*, New Age International Publishers, p. 298.

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Kruskal-Wallis: Solution and Conclusion

- Step 1 - Combine all 20 scores and rank them from lowest (1) to highest (20); no ties occur.

- $R(A) = \text{sum of ranks of } 271, 282, 257, 248, 262 = 14 + 18 + 7 + 4 + 10 = 53$

- $R(B) = \text{sum of ranks of } 252, 275, 302, 268, 276 = 5 + 15 + 20 + 12 + 16 = 68$

- $R(C) = \text{sum of ranks of } 260, 255, 239, 246, 266 = 9 + 6 + 1 + 3 + 11 = 30$

- $R(D) = \text{sum of ranks of } 279, 242, 297, 270, 258 = 17 + 2 + 19 + 13 + 8 = 59$

- Step 2 - Check: $53 + 68 + 30 + 59 = 210 = N(N+1)/2 = 20 \times 21 / 2$ ✓ (each $n_i = 5$, $N = 20$)

- $H = [12 / (20 \times 21)] \times \{53^2 / 5 + 68^2 / 5 + 30^2 / 5 + 59^2 / 5\} - 3 \times 21$

$$= 0.02857 \times 2362.8 - 63 = 67.51 - 63 = 4.51$$

$$df = k - 1 = 4 - 1 = 3.$$

Table value of $\chi^2(3)$ at $\alpha = 0.05$ is 7.815. Since $H = 4.51$ is smaller than 7.815, we accept H_0 — the bowler performs equally well with all four balls.

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Thanks